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Conditional Cash Transfers: A HANK Approach

1. Model and Methods

1.1 Model

The setup is a standard New Keynesian model with incomplete markets and labor supply risk. Time is discrete and runs from $t = 0$ to infinity. There are four types of agents in the economy: households, firms, labor unions, and the government.

Households are heterogeneous (Aiyagari 1994), workers and firms join in a formal-informal setup, and the rest of the economy follows the textbook New Keynesian model (Galí 2015). There is no aggregate uncertainty and the economy starts in the deterministic steady state.

Households

Assume there exists a continuum $i \in [0, 1]$ of ex-ante identical households, but experiencing idiosyncratic shocks in labor productivity, labor sector, and discount factor.

Households enjoy a GHH utility over consumption $c_{i,t}$ and labor $h_{i,t}$ (Greenwood, Hercowitz, and Huffman 1988), defined as a CRRA utility $u(\tilde{c}_{i,t}) = u(c_{i,t} - v(h_{i,t}))$,

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta_{i,t}^t \frac{\tilde{c}_{i,t}^{1-\frac{1}{\gamma}}}{1-\frac{1}{\gamma}} \right], \quad \text{with } v(h) = \psi \frac{h^{1+\frac{1}{\varphi}}}{1+\frac{1}{\varphi}},$$

where γ is the elasticity of intertemporal substitution (EIS) and φ is the Frisch elasticity of labor supply.

The GHH specification assumes a zero income effect on hours worked, so that the hours decision is separable from the consumption-savings problem, which seems to be a reasonable approximation for informal labor markets in development countries.

A household faces labor section risk $s \in \{F, I, U\}$, idiosyncratic labor productivity shocks $e_{i,t}$ with $\mathbb{E}[e_{i,t}] = 1$, and discount factor heterogeneity $\beta_i \in \{\beta_L, \beta_H\}$, where $\delta_\beta := \beta_H - \beta_L > 0$ and there is ω_I impatient households. The labor supply risk classifies households as belonging to the formal or informal sector, or as unemployed.

Formal workers earn labor income $(1 - \tau_\ell)w_t e_{i,t} h_{i,t}$, where w_t is the wage rate, τ_ℓ is a fixed labor income tax rate. While, informal workers receive benefits $w_t^I e_{i,t} h_{i,t}$, where $w_t^I = \xi w_t$ and $\xi \in (0, 1)$ is the wage gap. All unemployed and eligible informal households earn the benefit T_t . Let $y_t(s, e)$ be this non-financial income,

$$y_t(s, e) = \begin{cases} (1 - \tau_\ell)w_t e, & \text{if formal;} \\ w_t^I e h_t + T_t \cdot \mathbb{1}[w_t^I e h_t < \bar{y}], & \text{if informal;} \\ T_t, & \text{if unemployed.} \end{cases}$$

We assume that formal wages are observable by the government, so that only some informal workers and all unemployed may be eligible to receive Bolsa Família (BF). We assume that informal households are eligible if $w_t^I e_{i,t} h_{i,t} < \bar{y}$ and calibrate this to match to the number of beneficiaries.

Households save into liquid assets $a_{i,t}$ with real return $1 + r_t = (1 + i_{t-1})/(1 + \pi_t)$, where i_t represents the nominal interest rate and π_t denotes inflation. They also receive dividends $d_{i,t}$ from firms, which we assume to be proportional to productivity $d_{i,t} = d_t \cdot e_{i,t}$. Their budget constraint at time t reads:

$$c_{i,t} + a_{i,t} = (1 + r_t)a_{i,t} + d_{i,t} + y_{i,t}(s, e).$$

Incomplete markets impose a borrowing constraint $a_{i,t} \geq \underline{a}$. Frictions in the labor market imply formal households delegate their labor supply decisions to labor unions, and thus take hours worked as given. While, informal ones choose hours to maximize their utility, thus, for each i ,

$$\psi h^{1/\varphi} = w^I e \quad \Leftrightarrow \quad h(e) = \left[\frac{w^I e}{\psi} \right]^\varphi.$$

This generates a productivity cutoff for informal workers of

$$w^I e \left(\frac{w^I e}{\psi} \right)^\varphi < \bar{y} \quad \Leftrightarrow \quad e < e^* = \frac{(\psi^\varphi \bar{y})^{\frac{1}{1+\varphi}}}{w^I}.$$

We solved the household problem using the Endogenous Grid Method (Carroll 2006). The Bellman Equation for each individual i is:

$$\begin{aligned} V_t(s, \beta, e, a) &= \max_{c, a'} \left\{ \frac{c^{1-\frac{1}{\gamma}} - 1}{1 - \frac{1}{\gamma}} + \beta \mathbb{E}_t [V_{t+1}(s', \beta', e', a' | \cdot)] \right\} \\ \text{s.t.} \quad c + a' &= (1 + r_t)a + d_t + y_t(s, e) \\ a' &\in [\underline{a}, \infty). \end{aligned}$$

The labor supply risk is assigned by transition probabilities $\mathbb{P}(s' | s)$, where formal households may go to informality with probability p_{fi} and they don't go directly to unemployment, while informal households may go to formality with probability p_{if} and may lose their job with probability p_{iu} . Unemployed households go to formality with probability p_{uf} and to informality with p_{ui} .

$$\Pi_s = \mathbb{P}(s' | s) = \begin{bmatrix} 1 - p_{fi} & p_{fi} & 0 \\ p_{if} & 1 - p_{if} - p_{iu} & p_{iu} \\ p_{uf} & p_{ui} & 1 - p_{uf} - p_{ui} \end{bmatrix}.$$

In equilibrium, there is a measure of F_t formal households, I_t informal households, U_t unemployed, and BF_t beneficiaries of the BF program. We set p_{fi} and p_{if} to be endogenously determined in steady state. Let Φ_t be the outside option of the informal sector relative to formal employment,

$$\Phi_t = \frac{w_t^I + T_t}{(1 - \tau_\ell)w_t} - 1,$$

which is determined by a logistic switching model such that

$$\sigma_t = \frac{1}{1 + \exp(-\eta_s \Phi_t)}, \quad p_{fi} = \lambda_s \sigma_t, \quad p_{if} = \lambda_s (1 - \sigma_t),$$

where $\lambda_s := p_{fi} + p_{if} > 0$ is the total sector mobility rate, and $\eta_s \geq 0$ is the sensitivity of this mobility change in response to Φ_t . If $\eta_s = 0$, $p_{fi} = p_{if} = \lambda_s/2$. The idea is to calibrate for the total quarterly sector transition hazard from PNAD, as well as (p_{iu}, p_{uf}, p_{ui}) .

In this approach, Π_s is held fixed at its steady-state value, a short-run restriction to the intensive margin, treating the sector recomposition as a medium-run phenomenon.

Firms

In the firm side, there is a representative final-good producer that aggregates a continuum of $j \in [0, 1]$ intermediate inputs produced by formal firms,

$$Y_t = \left(\int_0^1 Y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

where $\varepsilon > 0$ is a constant substitution elasticity (CES).

Each intermediate good j is produced by a monopolistic competitive firm $y_{j,t} = Z_t l_{j,t}$. They hire labor at wage w_t and set a constant markup of $\mu = \frac{\varepsilon}{\varepsilon-1}$. Hence, $P_t = \mu W_t$, and the real wage equals $w_t = \frac{1}{\mu}$ at every date in steady state. Firms' dividends are taxed at the same rate τ_ℓ as labor income, so $d_t = (1 - \tau_\ell)(Y_t - w_t L_t)$, where $L_t = \mathbb{E}[l_{j,t}]$ is the aggregate formal labor.

Nominal wages are assumed to be sticky as in Erceg, Henderson, and Levin (2000), where unions set nominal wages to maximize agent utility subject to adjustment costs. We assume that unions allocate all formal labor hours uniformly across agents, so $\ell_{i,t} = L_t$ (Auclert, Rognlie, and Straub 2024).

Also, as in Hagedorn, Manovskii, and Mitman (2019), union's objective is to maximize the utility of an agent with average consumption $C_t = \int_0^1 c_{it} di$, discounted by the average discount factor $\bar{\beta}$, with Rotemberg (1982)'s adjustment costs. Thus results in a Phillips curve for price inflation is given by:

$$\ln(1 + \pi_t) = \kappa \left[\frac{w_t}{Z_t} - \frac{1}{\mu} \right] + \frac{Y_{t+1}}{Y_t} \cdot \frac{\ln(1 + \pi_{t+1})}{1 + r_{t+1}},$$

and for wage inflation $\pi_t^w = (1 + \pi_t)w_t/w_{t-1}$, we add a outside-option pass-through effect generated by differences between the formal and the informal sectors.

$$\ln(1 + \pi_t^w) = \kappa_w \left[\frac{w_t}{Z_t} - \frac{1}{\mu} + \phi_{out}\Phi_t \right] + \frac{\bar{\beta} \ln(1 + \pi_{t+1}^w)}{1 + r_{t+1}}.$$

Finally, there is a representative informal firm operating in a perfectly competitive market and producing under constant returns to scale, so $Y_t^I = w_t^I L_t^I$. The steady state wage gap $\xi = w_t/w_t^I$ is calibrated to match the PNAD data.

Government

Monetary Policy follows a standard lagged Taylor Rule, without microfoundations and only with an inflation target, $i_t = r_{t-1}^* + \phi\pi_{t-1}$, where r^* is the neutral real interest rate, and $\phi > 1$ (Taylor 1993).

In the governmental sphere, we have a government that purchase public goods, collects taxes, makes transfers, and incurs debt; therefore, the government's budget constraint is characterized by

$$G_t + T_t \cdot \text{BF}_t + (1 + r_{t-1})B_{t-1} = B_t + \tau_t w_t L_t,$$

which means that, in steady state, $G = \tau w L - r B - T \cdot \text{BF}$.

For dynamic simulations, we intend to apply two regimes: one financed by debt, with T fixed and $B_t = (1 + r)B + T \cdot \text{BF} - \tau w L - G$, and another financed by taxes, with B fixed and T variable. For now, we consider fixed T_t and B_t , treated as exogenous, with G_t adjusting to it.

Market Clearing

Finally, the asset, labor, and goods markets reach equilibrium (Market Clearing Conditions). We define $\Lambda_t(s, \beta, e, a)$ as the household distribution over a given period of time. First, the asset market reaches equilibrium when household savings equal government

bonds

$$A_t = \int a_{i,t} d\Lambda_t(s, \beta, e, a) = B_t;$$

the labor market clears when the share of employment in the workforce equals the firm's demand for labor

$$L_t = \int \mathbb{1}\{s_{i,t} = F\} d\Lambda_t(s, \beta, e, a) = F_t;$$

$$L_t^I = \int \mathbb{1}\{s_{i,t} = I\} d\Lambda_t(s, \beta, e, a) = I_t;$$

$$\text{and} \quad F_t + I_t + U_t = 1;$$

the BF benefit clears when it is given to the eligible households

$$\text{BF}_t = U_t + \int \mathbb{1}\{s_{i,t} = I, e_{i,t} < e^*\} d\Lambda(s, \beta, e, a);$$

and the goods market clear when $Y + Y^I = \bar{C} + G$ in the steady state. However, due to the GHH utility,

$$\int \mathbb{1}\{s_{i,t} = I\} \cdot v(h) d\Lambda(s, \beta, e, a) = \frac{\varphi w_t^I L_t^I}{1 + \varphi} = \frac{\varphi}{1 + \varphi} Y_t^I.$$

Thus,

$$Y_t + \frac{Y_t^I}{1 + \varphi} = \int c_{i,t} d\Lambda_t(s, \beta, e, a) = C_t + G_t.$$

1.2 Calibration

The main method is the Sequence-Space Jacobian (SSJ) (Auclert, Bardóczy, et al. 2021). The central idea is to summarize the model by a matrix equation $\mathbf{F}_t(\mathbf{X}, \mathbf{Z}) = \mathbf{0}$, where $\mathbf{X}$ are the endogenous variables and $\mathbf{Z}$ are the exogenous ones.

The method is built to find General Equilibrium Jacobians, written as

$$d\mathbf{X} = \mathbf{G} d\mathbf{Z} = -\mathbf{F}_\mathbf{X}^{-1} \mathbf{F}_\mathbf{Z} d\mathbf{Z}.$$

We are interested in calculating $dC = \mathbf{G}_{C,T} dT$, the total effect of the change in the conditional transfer on household consumption.

We resolve the heterogeneity by the Endogenous Grid Method (EGM) (Carroll 2006), which solves the value function iteration using the Euler equation, without solving the family maximization problem.

The idiosyncratic productivity follows a log-normal $AR(1)$ process, $\ln e' = \rho_e \ln e + \varepsilon^e$, with $\varepsilon^e \sim N(0, \sigma_e^2)$. This is approximated by a finite Markov chain with N_e grid points (Rouwenhorst 1995), which is more accurate than Taucher for highly persistent processes.

We also assume a log-spaced grid for assets, in order to generate more robustness near the credit constraint.

The discount factor transition (β_L, β_H) is modeled by a redrawing probability of $q = 0.10$, corresponding to an average type duration of 25 years, used to capture a generation (Auclert, Rognlie, and Straub 2025). This aims to capture similar results in relation to a life-cycle overlapping generations with permanent ability shock model.

We solve the steady state equilibrium computationally by Brent’s method, internally calibrating the parameters $(\beta_H, Z, \xi, G, p_{fi}, p_{if})$ in order to clear the asset market, the labor markets, the government budget constraint, and the equilibrium between sectors. The external calibration is shown in table 1. The goods market was left untargeted to verify the Walras’ Law.

2. Results

We solve the model at the calibrated steady state and compute both partial-equilibrium (PE) and general-equilibrium (GE) Jacobians with respect to the BF Transfer T , treating it as the fiscal instrument.

Steady state distribution. The stationary wealth distribution exhibits a sharp concentration near the borrowing constraint. A big number of households hold zero liquid assets and are therefore hand-to-mouth (HtM), while the Lorenz curve confirms the 2019 Survey of Consumer Finances shows for the US economy. These two features, a non-trivial HtM share and a sizeable pool of informal households, are the key distributional facts that make fiscal targeting welfare-relevant in this economy.

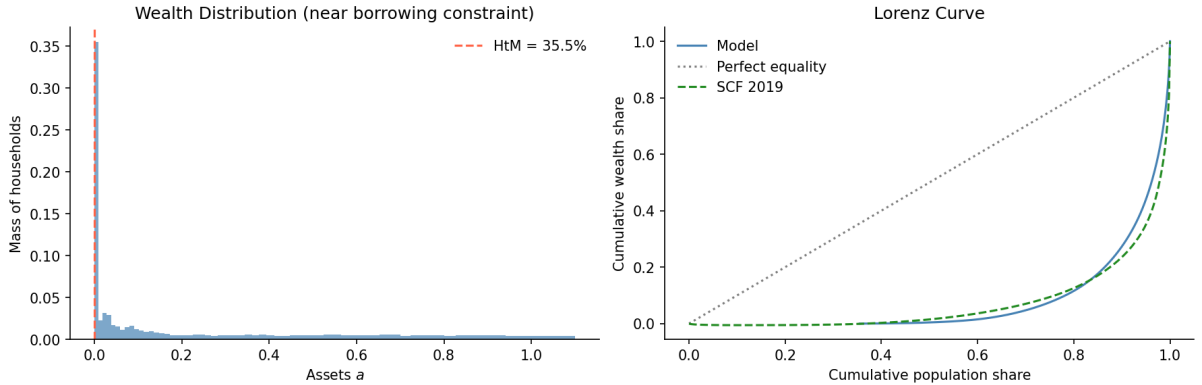


Figure 1: Wealth Distribution

Consumption policy function. The consumption policy function is nearly linear in assets for both employed and unemployed households at the median productivity level. The two schedules are nearly parallel, with the informal and the unemployed consuming slightly less at every asset level due to the lower benefit income. The slope of both curves

Table 1: Calibration

Parameter	Description	Value	Source / Target
<i>Households</i>			
γ	Elasticity of Intertemporal Substitution	0.5	Standard in HANK literature
ψ	Disutility of Labor	1.0	Standard in literature
φ	Frisch Elasticity	0.5	Standard in literature
β_H	Discount factor for the Patient	0.802	Calibrated: Asset Market
β_L	Discount factor for the Impatient	0.742	$\delta_\beta = 0.05$
ω_I	Share of Impatient Agents	0.25	Bartal and Becard (2024)
<i>Labor market</i>			
F	Formal sector size	0.423	$\lambda_s = 0.25, \eta_s = 3.0$
I	Formal sector size	0.433	$\lambda_s = 0.25, \eta_s = 3.0$
U	Unemployed Rate	0.144	$p_{iu} = 0.05, p_{ui} = 0.10, p_{uf} = 0.05$
<i>Idiosyncratic income process</i>			
ρ_e	Persistence of log-productivity	0.966	Kaplan, Moll, and Violante (2018)
σ_e	Std. dev. of productivity innovation	0.85	Bartal and Becard (2024)
n_e	Number of Rouwenhorst nodes	15	Standard discretization
<i>Asset grid</i>			
$\underline{a}$	Borrowing constraint	0.0	No borrowing (McKay and Reis 2016)
$\bar{a}$	Maximum asset	200	Sufficient to cover top of dist.
n_a	Number of asset grid points	200	Accuracy near constraint
<i>Prices and aggregate</i>			
Y	Aggregate Output	1.0	Normalization
Y^I	Informal Output	1.0	Normalization
Z	Total Productivity	2.37	Calibrated: Formal Labor Market
w	Real wage	2.13	Calibrated: $w = Z/\mu$
ξ	Wage gap $= w/w^I$	0.689	Calibrated: Informal Labor Market
<i>Government</i>			
τ_ℓ	Flat labor income tax rate	0.27	Kaplan, Moll, and Violante (2018)
$\bar{y}$	Bolsa Família Threshold	0.60	BF = 0.316
T	Bolsa Família Transfer	0.17	Shock variable of interest
B	Government Bond Supply	1.2	$B/Y = 120\%$
G	Government Spending	0.154	Calibrated: Government Budget
<i>Monetary policy</i>			
r^*	Neutral real interest rate	3.0%	12% annual average (BRL Data)
ϕ	Taylor rule inflation coefficient	1.5	Standard (Taylor 1993)
<i>Firms</i>			
μ	Price markup	1.11	10% markup (Basu and Fernald 1997)
κ	NKPC slope (Rotemberg)	0.1	Hazell et al. (2022)
ϕ_{out}	Pass-through coefficient between sectors	1.0	xxx

$\gamma = 0.5$ is consistent with micro estimates surveyed by Valickova, Havranek, and Horvath (2015).

is close to the real interest rate r , indicating that high-asset households are close to permanent-income consumers and will save most of any windfall.

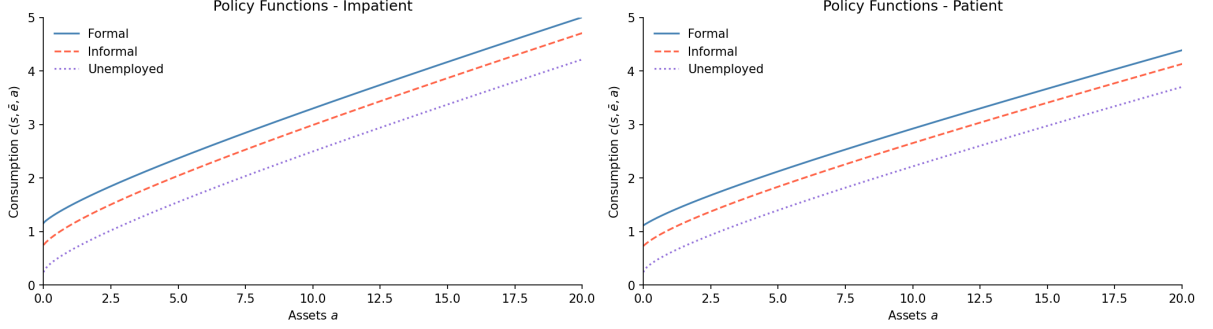


Figure 2: Steady State Consumption Policy Function

Intertemporal MPC profile. The PE intertemporal MPC of aggregate consumption with respect to T decays rapidly. This profile reflects the fact that informal households are disproportionately constrained and spend their additional income quickly. The monotone decay, without the hump shape typical of models with patient savers, is consistent with the one-asset structure: there is still no liquid-illiquid wedge that would delay consumption.

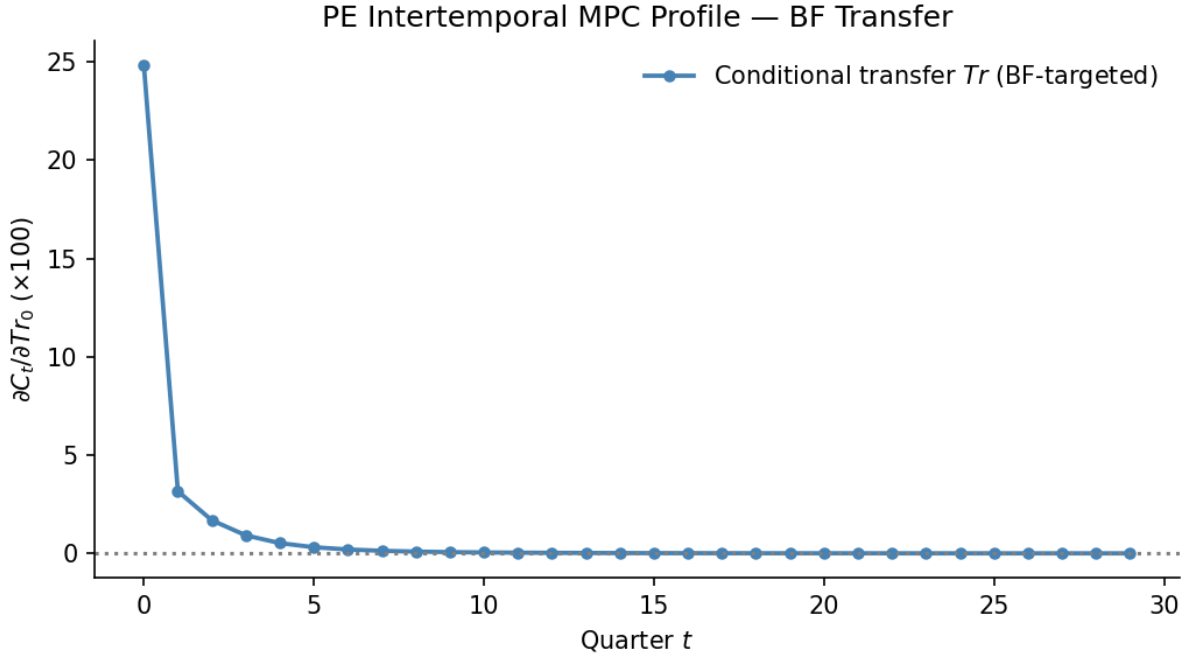


Figure 3: Partial Equilibrium MPC Profiles

General equilibrium impulse responses. We compute GE impulse responses to a persistent 1% rise in T (AR(1) with $\rho = 0.4$). The GE output and consumption multipliers coincide with the PE impact, indicating that GE amplification through wage and interest-rate channels is negligible for this shock at the current calibration. The unemployment

rate IRF is essentially zero, since employment transitions are still exogenous in this model, and the GE wage response is too small to generate meaningful labor-market feedback.

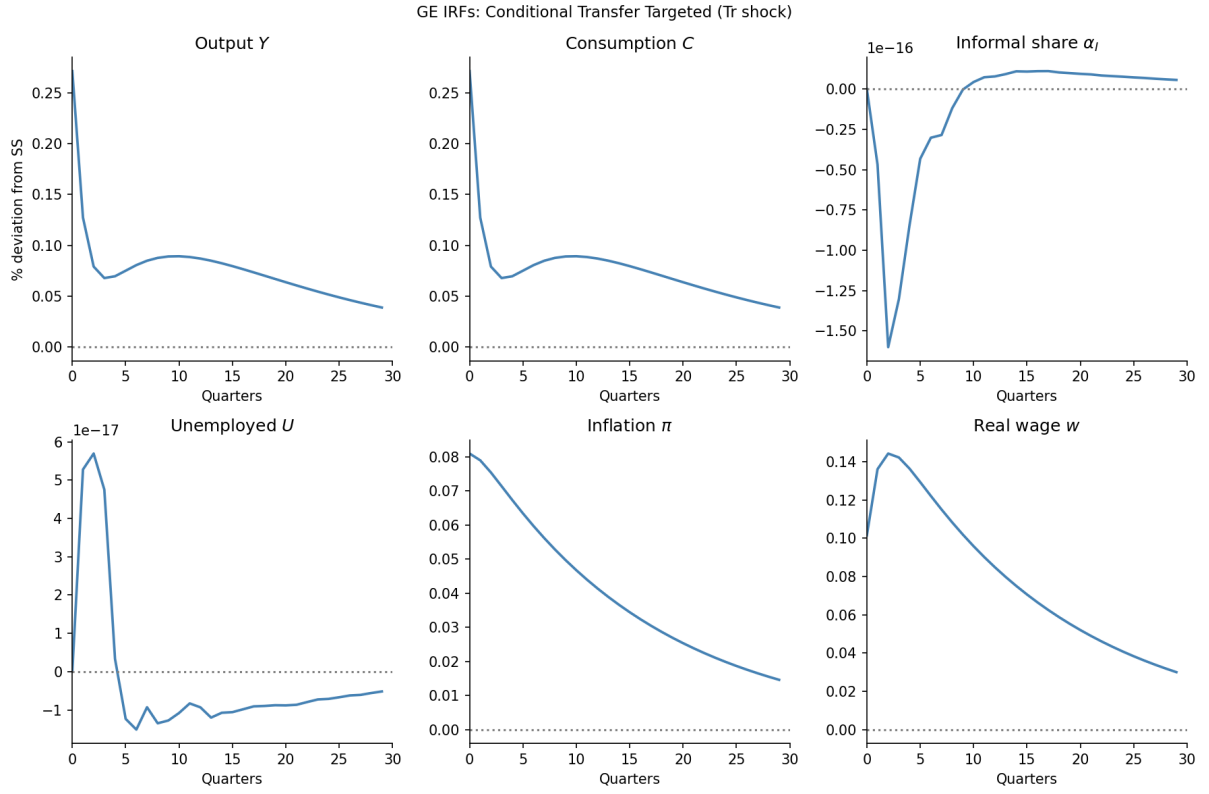


Figure 4: IRFs: BF-Targeted (shock in T)

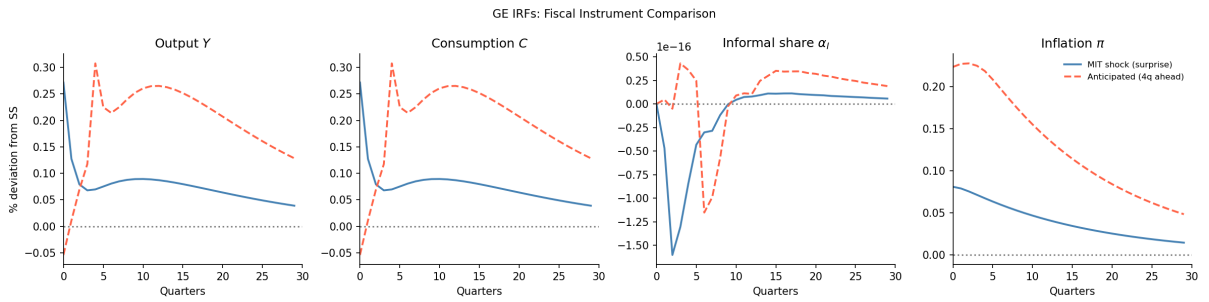


Figure 5: IRFs: Comparison (shock in T , MIT vs Anticipated)

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